Skill-20

```
himasree@LAPTOP-41COT5AA:/mnt/c/Windows/system32$ cd ~/OSLab
himasree@LAPTOP-41COT5AA:~/OSLab$
himasree@LAPTOP-41COT5AA:~/OSLab$ nano sigstp_jobs.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -Wextra process_groups.c -o process_groups
cc1: fatal error: process_groups.c: No such file or directory
compilation terminated.
himasree@LAPTOP-41COT5AA:~/OSLab$ nano process_groups.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -Wextra sigstp_jobs.c -o sigstp_jobs
himasree@LAPTOP-41COT5AA:~/OSLab$ ./sigstp_jobs
Parent PID: 592
Parent process group ID: 592
Child PID: 593
Child process group ID: 593
Child running for 5 seconds...
Child finished.

[1]+  Stopped                 ./sigstp_jobs
himasree@LAPTOP-41COT5AA:~/OSLab$ kill -TSTP PARENT_PID
-bash: kill: PARENT_PID: arguments must be process or job IDs
himasree@LAPTOP-41COT5AA:~/OSLab$ ./sigstp_jobs
Parent PID: 628
Parent process group ID: 628
Child PID: 629
Child process group ID: 629
Child running for 5 seconds...
kill -TSTP PARENT_PID
```

```
himasree@LAPTOP-41COT5AA:/mnt/c/Windows/system32$ cd ~/OSLab
himasree@LAPTOP-41COT5AA:~/OSLab$ nano process_groups.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc -Wall -Wextra process_groups.c -o process_groups
himasree@LAPTOP-41COT5AA:~/OSLab$ ./process_groups
Parent PID: 624
Child PID: 625
Parent process group ID: 624
Child process group ID: 625
Child running for 5 seconds...
Child finished.

[1]+  Stopped                 ./process_groups
himasree@LAPTOP-41COT5AA:~/OSLab$
```

```c
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <signal.h>
#include <sys/wait.h>
#include <termios.h>

int main()
{
    pid_t pid;
    pid_t shell_pgid;
    int status;
    struct termios shell_modes;

    if (!isatty(STDIN_FILENO))
    {
        printf("Run this program in a terminal.\n");
        return 1;
    }

    shell_pgid = getpgrp();
    tcgetattr(STDIN_FILENO, &shell_modes);

    pid = fork();
```

[Read 70 lines]

```c
    else
    {
        setpgid(pid, pid);

        printf("Parent PID: %d\n", getpid());
        printf("Parent process group ID: %d\n", getpgrp());

        if (tcsetpgrp(STDIN_FILENO, pid) == -1)
        {
            perror("tcsetpgrp");
            kill(pid, SIGTERM);
            waitpid(pid, NULL, 0);
            return 1;
        }

        waitpid(pid, &status, WUNTRACED);

        tcsetpgrp(STDIN_FILENO, shell_pgid);
        tcsetattr(STDIN_FILENO, TCSADRAIN, &shell_modes);

        printf("Terminal control restored to parent.\n");
    }

    return 0;
}
```

```c
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <signal.h>
#include <sys/wait.h>
#include <termios.h>

int main()
{
    pid_t pid;
    pid_t shell_pgid;
    int status;
    struct termios shell_modes;

    if (!isatty(STDIN_FILENO))
    {
        printf("Run this program in a terminal.\n");
        return 1;
    }

    shell_pgid = getpgrp();
    tcgetattr(STDIN_FILENO, &shell_modes);

    pid = fork();
```

```c
    {
        setpgid(pid, pid);

        printf("Parent PID: %d\n", getpid());
        printf("Parent process group ID: %d\n", getpgrp());

        if (tcsetpgrp(STDIN_FILENO, pid) == -1)
        {
            perror("tcsetpgrp");
            kill(pid, SIGTERM);
            waitpid(pid, NULL, 0);
            return 1;
        }

        waitpid(pid, &status, WUNTRACED);

        tcsetpgrp(STDIN_FILENO, shell_pgid);
        tcsetattr(STDIN_FILENO, TCSADRAIN, &shell_modes);

        printf("Terminal control restored to parent.\n");
    }

    return 0;
}
```